3.2.1 Co-efficient of a term

A term of an algebraic expression is a product of factors. Here each factor or product of factors is called the **co-efficient** of the remaining product of factors.

For example, in the term $5xy$, 5 is the co-efficient of remaining factor product xy. Similarly x is the co-efficient of $5y$; $5x$ is the co-efficient of y. The constant 5 is called the **numerical co-efficient,** and others are called simply co-efficients.

A co-efficient can either be a numerical factor or an algebraic factor or product of both.

Since we often talk about the numerical co-efficients of a term, if we say "co-efficient", it will be understood that we are referring to the numerical co-efficient. If no numerical co-efficient appears in a term, then the co-efficient is understood to be 1.

Consider the term $-6ab$. It is the product of three factors -6, a and b. Also it can be written as a product of two factors such as $-6a \times b$, $-6b \times a$ and $-6 \times ab$.

The co-efficient of 'a' is $-6b$

The co-efficient of 'b' is $-6a$

The co-efficient of 'ab' is -6

Thus -6 is the **numerical co-efficient** in the term $-6ab$.

Example 3.2 Find the numerical co-efficient of the following terms. Also, find the co efficient of x and y in each of the term: $3x$, $-5xy$, $-yz$, $7xyz$, y, $16yx$.

Solution

Term	Numerical co-efficient	Co-efficient of x	Co-efficient of y
$3x$	3	3	Not possible
$-5xy$	-5	$-5y$	$-5x$
$-yz$	-1	Not possible	$-z$
$7xyz$	7	$7yz$	$7xz$
y	1	Not possible	1
$16yx$	16	$16y$	$16x$

3.3 Like and unlike terms

When you visit a market, you can see that, the vegetables and fruits of same kind are kept as separate heaps. Similarly, we can group the same kind of terms in an algebraic expression.



*Fig.*3.3



*Fig.*3.4

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